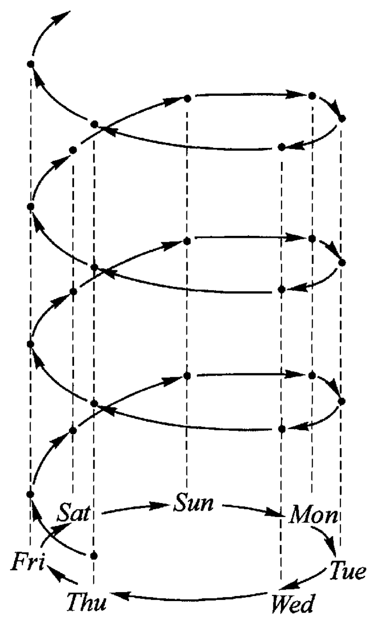

Please write your name in the box below.

Basic Algebra I Midterm 2

21 February 2025, 12.30 PM–1.20 PM

Weight	*Problem*
5	*1(i)*
10	*1(ii)*
5	*2(i)*
10	*2(ii)*
10	*2(iii)*
10	*3*
50	

Problem 1.

(i) What does it mean to say that a subset $H \subset G$ of a group is a *subgroup*?

(ii) Let $\alpha : X \to X$ be a function from a set to itself. Show that[1]

$$\mathrm{Cent}(\alpha) = \{\sigma \in \mathrm{Sym}(X) : \sigma \circ \alpha = \alpha \circ \sigma\}$$

is a subgroup of $\mathrm{Sym}(X)$.

[1]Recall that $\mathrm{Sym}(X)$ denotes the set of bijective functions $X \to X$, regarded as a group under the binary operation of composition.

Problem 2. Let G and G' be groups.

(i) What does it mean to say that a function $\phi : G \to G'$ is a *group homomorphism*?

(ii) Suppose now that $\phi : G \to G'$ is a group homomorphism, and that $H' \subset G'$ is a subgroup. Show that $\phi^{-1}(H')$ is a subgroup of G.

(iii) With notation as in (ii), show for every sequence of elements $g_1, \ldots, g_r \in G$ that[2]

$$\phi(\langle g_1, \ldots, g_r \rangle) \subset H' \qquad \Leftrightarrow \qquad \phi(g_1), \ldots, \phi(g_r) \in H'.$$

[2]Recall that when $k_1, \ldots, k_r \in K$ are elements of a group, $\langle k_1, \ldots, k_r \rangle$ is the subgroup uniquely characterized by the property: for every subgroup $K' \subset K$, one has

$$\langle k_1, \ldots, k_r \rangle \subset K' \qquad \Leftrightarrow \qquad k_1, \ldots, k_r \in K'.$$

Problem 3. Find an integer x satisfying

$$[x]_{15} = [8]_{15} \qquad \text{and} \qquad [x]_{29} = [24]_{29}.$$